MATH 101

Problem Set 4

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Sequences and Series

Let (a_n) be a sequence of real numbers. Prove that if $\lim_{n\to\infty} a_n = L$, then every subsequence of (a_n) also converges to L.

Proof. Let (a_{n_k}) be any subsequence of (a_n). Fix $\varepsilon > 0$. Since $a_n \to L$, there exists $N \in \mathbb{N}$ such that $|a_n - L| < \varepsilon$ for all $n \geq N$. Since $n_k \geq k$ for all k (the indices are strictly increasing), we have $n_k \geq N$ for all $k \geq N$, so $|a_{n_k} - L| < \varepsilon$. Hence $a_{n_k} \to L$. □

Let $\sum_{n=1}^{\infty} a_n$ be a convergent series of non-negative terms. Show that $\sum_{n=1}^{\infty} a_n^2$ also converges.

Proof. Since $\sum a_n$ converges, we have $a_n \to 0$, so there exists N such that $a_n < 1$ for all $n \geq N$. For such n, $a_n^2 \leq a_n$. The convergence of $\sum_{n\geq N} a_n^2$ then follows by comparison. □

Determine, with proof, whether the series

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$$

converges absolutely, conditionally, or diverges.

Proof. The series converges conditionally. By the alternating series test, since $1/n \searrow 0$, the series converges. However, $\sum 1/n$ diverges (harmonic series), so convergence is not absolute. □

Continuity

Let $f, g\colon \mathbb{R} \to \mathbb{R}$ be continuous. Define $h(x) \coloneqq \max\{f(x), g(x)\}$. Show that h is continuous.

Proof. Use the identity $\max\{a, b\} = \frac{1}{2}(a + b + |a - b|)$. Since f and g are continuous, so are $f + g$ and $f - g$, and hence $|f - g|$ (as the composition of continuous functions). Therefore $h = \frac{1}{2}(f + g + |f - g|)$ is continuous. □

Exercise 0.5. State and prove the analogous result for $\min\{f(x), g(x)\}$.

Bonus. Give an example of a function $f\colon \mathbb{R} \to \mathbb{R}$ that is continuous at exactly one point. ✍